

## Research Article

# Early psychosocial interventions after disasters, terrorism, and other shocking events: Guideline development

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## Abstract

Although most victims of disasters, terrorism, or other shocking events recover on their own, a sizable amount of these victims develops long-term disaster-related problems. These victims should receive timely and appropriate psychosocial help. This article describes the development of guidelines on psychosocial interventions during the first 6 weeks after a major incident. Scientific literature, expert opinions, and consensus among relevant parties in the clinical field were used to formulate the recommendations. Early screening, a supportive context, early preventive and curative psychosocial interventions, and the organization of interventions are covered. The implications for the clinical field and future research are discussed. It is concluded that the international knowledge base provides valuable input for the development of national guidelines. However, the successful implementation of such guidelines can take place only if they are legitimated and accepted by local key actors and operational target groups. Their involvement during the development process is vital.

## Key words

disaster medicine, disaster planning, post-traumatic stress disorders, practice guidelines, social support, terrorism care.

## INTRODUCTION

Disasters, terrorism, and other shocking events have a profound effect on people's lives. Although research has shown that most victims of such events recover on their own, a sizable amount of these victims develops long-term disaster-related problems (Norris *et al.*, 2002). Recent disasters in the Netherlands have shown that the psychosocial and health effects of a disaster sometimes last for years. This underscores the necessity for timely and appropriate psychosocial help for victims of disasters.

In the Netherlands, after shocking events, acute psychological trauma support is offered by a variety of caregivers (i.e. professionals and volunteers, non-profit and commercial organizations). Notwithstanding the variety of people providing psychosocial care, the care itself should be offered in accordance with the latest practices that have been shown to be effective. However, on various issues, there is a discrepancy between the results of scientific studies into the effectiveness of early psychosocial interventions and what actually happens in practice. There are debates about the time when early psychosocial interventions should take place, who should carry them out to greatest effect, and which methods

ought to be followed. Answers to questions like these often depend on the views of organizations, individual relief workers, and sometimes on commercial interests. As a result, after-care is often given in many different ways and the victims are not always provided with “state of the art” care.

The professional community is in need of clarity. As a response, in the Netherlands, evidence-based multidisciplinary guidelines on early psychosocial interventions after disasters, terrorism, or other shocking events were developed. Guidelines are a means of providing help in accordance with the latest insights from science and based on (systematically collected) practical experience. They provide answers to the question of “What (not) to do” during the first 6 weeks after a disaster, terrorist attack, or other shocking event. A period of 6 weeks was chosen because: (i) during this period, most early interventions are offered; and (ii) it is known that, if stress reactions disappear “by themselves”, it will be during the first 6 weeks. The aim of this article is to summarize the procedure by which this Dutch standard for psychosocial care was developed and to give an overview of the resulting guidelines.

## METHODS

A multidisciplinary national panel of experts (hereafter dubbed “The Panel”) was formed to develop the guidelines, consisting of 21 members. The Panel represented the five main associations of Dutch health-care professionals and

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**Table 1.** Classification of the methodological quality of individual studies

| Level | Intervention                                                                                                                        | Diagnostic accuracy of the study                                                                                                                                                                                                                                                                    | Damage or side-effects, etiology, and prognosis                                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1    | Systematic review of at least two studies of A2 level carried out independently of each other                                       |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                       |
| A2    | Randomized, double-blind comparative clinical study of good quality and of sufficient size                                          | Study in relation to a reference test (a “gold standard”) with previously defined cut-off points and an independent assessment of the results of the test and the gold standard, relating to a sufficiently large series of consecutive patients, all of whom have had the index and reference test | Prospective cohort study of sufficient size and follow-up in which there has been adequate checking for “confounding” factors and selective follow-up has been sufficiently ruled out |
| B     | Comparative study, but not with all the characteristics of those of A2 (this also includes patient control studies, cohort studies) | Study in relation to a reference test, but not with all the characteristics of A2                                                                                                                                                                                                                   | Prospective cohort study, but not with all the characteristics of A2, or a retrospective cohort study or patient control study                                                        |
| C     | Non-comparative study                                                                                                               |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                       |
| D     | Expert opinion                                                                                                                      |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                       |

13 additional associations and organizations specifically involved in early psychosocial interventions (see Appendix I for an overview of these organizations). Starting questions were formulated based on the obstacles experienced in day-to-day postdisaster care. These questions were answered using relevant scientific studies along with additional, more practical considerations by members of The Panel. Using this methodology, the conclusions that are drawn can be considered “evidence-based”. Although the guidelines are aimed at psychosocial care after major disasters, the literature used was primarily based on smaller-scale events. The Panel considered this literature as suitable for these guidelines, given the comparability of the effects of small-scale and large-scale events and the absence of studies showing the opposite.

Systematic literature searches were carried out, focused on publications between 1995 and 2006 (supplemented by “key articles” from before 1995). The literature was drawn from a number of sources: the National Guidelines Clearinghouse and Guideline International Network, The Cochrane database of “systematic reviews” of the Cochrane Library up to 2006, Medline (PubMed), PsycINFO, and Pilots. The search focused on existing evidence-based guidelines for early post-disaster psychosocial interventions and systematic reviews or meta-analyses, using search terms such as “disasters”, “terrorism”, “acute post-traumatic stress”, “acute psychological interventions”, “crisis care”, “brief interventions”, and “debriefing”. In addition, relevant articles from the bibliographies of selected articles were obtained.

The guidelines that were found were assessed for the quality of the methodology by using the AGREE instrument (The AGREE Collaboration, 2001). The quality of the selected articles was first classified based on their methodological design, ranging from level A to level D (see Table 1). The classification of methodological quality, as depicted in Table 1, is not entirely applicable to studies into psychologi-

**Table 2.** Level of evidence of conclusions

| Conclusion | Basis                                                                                      |
|------------|--------------------------------------------------------------------------------------------|
| 1          | Level A1 study or at least two level A2 studies carried out independently of each other    |
| 2          | One level A2 study or at least two level B studies carried out independently of each other |
| 3          | One level B or C study                                                                     |
| 4          | Expert opinion                                                                             |

cal interventions because these cannot be carried out double-blind. Therefore, single-blind, randomized controlled trials were assigned an A level also. Next, the evidence level was determined, ranging from level 1 to level 4 (see Table 2).

The “evidence-based” conclusions were based on the available evidence from the literature and additional considerations by The Panel. These conclusions were followed by recommendations. In order to arrive at a recommendation, additional aspects were taken into consideration, including the preferences of victims, costs, availability, and organizational aspects. Unfortunately, some starting questions could not be answered satisfactorily due to a lack of unanimous evidence. For these topics, The Panel made recommendations for additional research.

## CONTENTS OF THE GUIDELINES

The contents of the guidelines were divided into five topics: (i) screening; (ii) supportive context; (iii) preventive early interventions; (iv) curative preventive interventions; and (v) organization. In the following section, the conclusions of The Panel are summarized, accompanied by their specific evidence level. Where relevant, the resulting recommendations are also described (an overview of all recommendations is found in Table 3).

**Table 3.** Recommendations made by The Panel

| No. | Recommendation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | With regard to the aims of early psychological interventions:<br>Early psychosocial interventions serve the following purposes: (i) the stimulation of natural recovery and the use of natural resources; (ii) the identification of those affected by a disaster, terrorism, or other shocking events who are in need of acute psychosocial care; and (iii) if necessary, to refer and treat those affected and in need of acute psychosocial care                                                                                                                                                                                                                                                                                                                                                                                     |
| 1   | With regard to screening:<br>Despite some evidence, no early tracing of those affected who have a high risk of a PTSD by using PTSD questionnaires                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2   | Screening instruments that are not in Dutch (usually English) should be translated and then validated among those affected who are Dutch-speakers; validation must be carried out in the Dutch populations of those affected by disasters, terrorism, or other shocking events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3   | In the case of pre-existing mental disorders, proper diagnosis and treatment should be carried out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4   | No early tracing of those affected who have a high risk of a PTSD by using ASD as a predictor; however, planning a follow-up meeting for further observation is recommended                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5   | Further studies of the populations of those affected by disasters, terrorism, or other shocking events into the usability of screening instruments that are based on risk factors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6   | Further studies into the effectiveness of population-wide screening after traumatic events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7   | After a shocking event, if it is decided to screen children and adolescents for symptoms of ASD, information should be gathered from both the child and the parents/caregivers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8   | Further studies into the most appropriate time for screening after a disaster, terrorism, or other shocking event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 9   | Further studies into the potentially negative effects of the screening of victims of disasters, terrorism, and other stressful life events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 10  | With regard to the supportive context:<br>After a disaster, terrorism, or other shocking event, a supportive context is offered that consists of: (i) listening, support, solace, and being open to the immediate practical needs of those affected; (ii) factual and up-to-date information about the shocking event; (iii) the mobilization of social support from the social surroundings of those affected; (iv) reunification with the people closest to those affected and keeping families together; and (v) reassurance for those affected who display normal stress reactions                                                                                                                                                                                                                                                  |
| 11  | With regard to early preventive psychosocial interventions:<br>After a disaster, terrorism, or other shocking event, information is offered to all those affected, which should consist of: (i) a reassuring explanation about normal reactions; (ii) knowledge of when to seek help; and (iii) advising those affected to tackle their daily routine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 12  | Do not offer preventive psycho-education                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 13  | In the acute phase, trained volunteers, and preliminary and primary relief workers, with secondary mental health-care workers in the background, have a role to play in the area of psychological triage in order to identify those affected who have mental disorders and/or serious clinical symptoms requiring diagnosis and/or treatment; psychological triage often takes place in the acute phase, but also can take place during the period to which this guideline relates (the first 6 weeks) (e.g. when a person who has been affected consults a caregiver)                                                                                                                                                                                                                                                                  |
| 14  | With psychological triage, a distinction must be made between: (i) those affected who do not have mental disorders and/or serious clinical symptoms (the largest group of those affected; most will recover within a foreseeable time on their own and in the first 6 weeks; if necessary, they only need to be reassured and be given a small amount of information); (ii) those affected and who might have mental disorders and/or serious clinical symptoms (a small amount of information also needs to be given to this group; in addition, the caregiver must arrange a follow-up meeting with the person affected); and (iii) those affected who have mental disorders and/or serious clinical symptoms (there are evident clinical problems in this group and proper diagnosis and/or treatment should be offered immediately) |
| 15  | The avoidance of once-only psychological debriefing (including CISM), with the aim of preventing a PTSD and other psychological problems in those affected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 16  | No psychological debriefing (including CISM) for members of the emergency services                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 17  | Support from colleagues (peer support) contributes to a supportive context as it can provide practical and emotional support and can encourage the use of the social resources of support of the person affected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 18  | No offering of CISM to a wide population                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 19  | Further studies into the effectiveness of CISM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 20  | Further studies into the effectiveness of the psychological first aid module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 21  | The avoidance of once-only psychological debriefing (including CISM), with the aim of preventing a PTSD and other psychological problems in affected children                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 22  | Further studies into the effectiveness of a structured relief protocol as a preventive intervention for children                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 23  | With regard to early curative psychosocial interventions:<br>Treatment with trauma-focused CBT for those affected who have an ASD or severe symptoms of PTSD in the first month after a shocking experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 24  | Relaxation should be offered only as part of CBT, not as a (non-trauma-focused) intervention on its own                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 25  | Further studies into the effectiveness of EMDR as a curative early intervention in the first 6 weeks after stressful life events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 26  | In the event of sleep disorders as a result of a trauma, pharmacotherapy may be considered (for any drug treatment for sleep disorders, depressive disorders, or anxiety disorders, the panel refers the reader to the existing guidelines)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 27  | The employer should offer counseling (to be carried out by a relief worker or trained volunteer) if a shocking event takes place at work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 28  | Treatment with trauma-focused CBT for children > 7 years of age with severe symptoms of acute post-traumatic stress and/or an ASD in the first month after a shocking event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 29  | Further studies into the effectiveness of EMDR as an early curative intervention in children                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 30  | Further studies into pharmacological interventions in children                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 31  | Approach ethnic minorities as regularly as possible and as culture-specifically as necessary: culture-specific elements may consist of giving information in the mother tongue and involving key figures from the ethnic minority group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 32  | With regard to the organization of early psychosocial care:<br>Early psychosocial interventions should be carried out by people who are trained/have been given special instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 33  | Collective early interventions form an essential part of the integrated package of postdisaster psychosocial care; therefore, the use of these interventions should be ensured                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 34  | In the first 6 weeks after a disaster, a good support system should be set up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 35  | Information is given a fixed place in the various policy plans in order for this to run smoothly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 36  | Profession-specific implementation of this generic multidisciplinary guideline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

ASD, acute stress disorder; CBT, cognitive-behavioral therapy; CISM, critical incident stress debriefing; CISM, critical incident stress management; EMDR, eye movement desensitization and reprocessing; PTSD, post-traumatic stress disorder.

The scientific literature is not univocal about the aims of early psychosocial interventions after disasters. The Panel acknowledges that early psychosocial interventions should be given in order to ensure the mental well-being of those affected and to prevent psychopathology. Therefore, The Panel concluded that psychosocial interventions should achieve the following aims: (i) the promotion of natural recovery and the use of natural resources; (ii) the identification of victims in need of acute psychological help; and (iii) the referral and treatment of victims who need acute psychological help (evidence level 4).

### Screening (recommendations 1–9)

It is important to differentiate screening from triage and detection. Screening means to look systematically in specific target populations for an illness that has not manifested yet. Triage, in contrast, is used to identify victims of a disaster situation who need acute care. Finally, detection is a broad term that aims at being alert to the possible presence of an illness, without specifically examining an entire target population. The Panel does not consider screening, on the basis of post-traumatic stress disorder (PTSD) questionnaires, to be beneficial. There are some promising results on early screening for PTSD (evidence level 2: Brewin *et al.*, 2002; Silove *et al.*, 2003), but these studies need to be replicated within populations of victims of disasters.

The literature does not agree on the predictive value of acute stress disorder (ASD) for PTSD (evidence level 3). In many cases, stress reactions during the first month are better understood as “normal reactions to abnormal events”. Also, further research is needed into the effectiveness of screening instruments based on risk factors. Although there are studies that show clear risk factors for PTSD, the predictive value of these is too small to be used in practice (evidence level 1: Brewin *et al.*, 2000; Ozer *et al.*, 2003; evidence level 3: Wohlfarth *et al.*, 2002; Winkel *et al.*, 2004).

The screening of acute stress syndrome among children has been done with some success (evidence level 3: Chemtob *et al.*, 2002). However, both parents and professionals in schools are often protective towards children, and understandably so. They are resistant to “outsiders” who want to obtain information about and from the children. The Panel recommends that, if after a shocking event it is decided to screen children and adolescents for symptoms of ASD, information should be gathered from both the child and the parents/caregivers.

The Panel also considered the possible negative effects of screening, such as the “medicalization” of people who suffer acute symptoms of stress as part of the natural process. Also, screening will produce false-positives and false-negatives. Furthermore, the costs and amount of organization associated with the screening of large groups of affected people probably do not correspond with the benefits in terms of health outcomes. Finally, more research is needed on the most appropriate time for when screening is needed.

Overall, The Panel recommends that, currently, screening during the first 6 weeks after a disaster is not advisable. Nevertheless, during this period, all efforts must be made to iden-

tify those victims in need of direct psychological help and proper diagnosis and treatment must be constantly available.

### Supportive context (recommendation 10)

There is a broad consensus on the value of offering a supportive context (evidence level 4). Offering solace, information, and support and attending to immediate practical and emotional needs play a valuable role in the acute coping of victims (Litz *et al.*, 2002). Practical and social support play an important part in recovering from PTSD symptoms, particularly directly after an incident (NICE, 2005). Based on these and other considerations (e.g. Parker *et al.*, 2006), The Panel recommends that a supportive context should be provided, consisting of: (i) listening, support, solace, and being open to the immediate practical needs of those affected; (ii) factual and up-to-date information about the incident; (iii) the mobilization of social support from the victims’ own social surroundings; (iv) reunification with the people closest to those affected and keeping families together; and (v) reassurance for those affected who display normal stress reactions.

An outreaching approach should be preferred, both in the acute phase and afterwards (i.e. relief workers should not wait until victims “bring their problems” on their own accord). The supportive context can be facilitated further by enabling victims to get into contact with others who have been affected or who have had similar experiences. These contacts can be organized in self-help groups or groups of people with similar experiences. For children, the provision of a supportive context must be provided through adults and a social network that are close to the child (usually the school or other existing [e.g. sports] organization).

The notion of a supportive context can be considered to be pivotal for these guidelines. It is the opinion of The Panel that a supportive context must be available and accessible at all times during the first 6 weeks. It forms the basis from which additional psychosocial interventions are carried out. Two types of early psychosocial interventions can be distinguished: preventive and curative interventions. These are described in the next subsections.

### Early preventive psychosocial interventions (recommendations 11–22)

The Panel distinguishes the following topics in preventive early interventions: (i) offering general information; (ii) psychological triage; (iii) psychological debriefing; and (iv) other early intervention methods. Considering the value of offering general information to victims, there is a broad consensus that this is needed (evidence level 4). Such information should consist of a reassuring explanation about normal reactions, the provision of the indications for when to seek help, and advice on how to continue with the daily routine. For children, the information should be customized to the child’s level of understanding and experience.

The provision of general information should not be confused with psycho-education. Psycho-education consists of structured (often repeated) information and training to those



affected. Its aim is to bring about change in trauma-related behavior. There is no scientific support for the effectiveness of such preventive psycho-education (evidence level 1: e.g. Ehlers *et al.*, 2003; NICE, 2005; Turpin *et al.*, 2005; Sijbrandij *et al.*, 2007). Therefore, The Panel does not recommend offering preventive psycho-education, as defined above.

Besides providing general information, in the acute phase, trained volunteers, and preliminary and primary relief workers have a role to play in psychological triage. Psychological triage consists of identifying victims with mental disorders and/or serious clinical symptoms requiring diagnosis and/or treatment. This implies that both volunteers and professionals need to have some training in order to recognize clinically evident problems. Evidence on the effectiveness of psychological triage is not available. In general, The Panel recommends that professionals, and others as well (e.g. the partners of victims, supervisors, and school staff), should use the first 6 weeks to observe the victims closely. When there are evident clinical problems, direct and accurate diagnosis and treatment should be offered (evidence level 4).

Psychological debriefing is a third preventive intervention. It can be described as a standardized crisis intervention, the purpose of which is to prevent and reduce the adverse psychological effects of traumatic events. Psychological debriefing takes many different forms, but it is often understood as a once-only, semistructured intervention. It has been shown that psychological debriefing after a shocking event is not effective in preventing PTSD and other psychological problems and that single-session debriefing can even have damaging effects (evidence level 1: e.g. Van Emmerik *et al.*, 2002; Lewis, 2003; Aulagnier *et al.*, 2004; Sijbrandij *et al.*, 2007). The evidence does not support the use of psychological debriefing for children either (Stallard *et al.*, 2006). With the aim of preventing PTSD and other psychiatric problems in those affected, The Panel recommends the avoidance of once-only psychological debriefing.

Other available early intervention methods are critical incident stress management (CISM) and "psychological first aid", a modular early intervention that was developed by The National Center for Post-traumatic Stress Disorder. There are no indications that CISM actually prevents the development of chronic psychological disorders (evidence level 1: e.g. Everly *et al.*, 2002; Roberts & Everly, 2006). The psychological first-aid module, although promising and scientifically well founded, has not yet been sufficiently proven to be effective in scientific studies. Therefore, The Panel recommends further studies into the effectiveness of both CISM and the psychological first aid module.

### **Early curative psychosocial interventions (recommendations 23–31)**

The guidelines on early curative interventions are limited to interventions that can be offered during the first 6 weeks after a disaster. The duration of treatment for most commonly occurring conditions postincident (e.g. anxiety disorders, depressive disorders, acute stress disorder, PTSD) extends well beyond 6 weeks. In addition, PTSD can only be diagnosed 4 weeks following a stressful life experience

(American Psychiatric Association, 1994). Furthermore, the treatments for these conditions are described extensively in existing Dutch multidisciplinary guidelines.

The combination of cognitive and behavioral therapeutic procedures is called cognitive-behavioral therapy (CBT). Research indicates that brief, trauma-focused CBT during the first few weeks after a shocking event leads to a reduction in PTSD symptoms a number of months later (evidence level 1: André *et al.*, 1997; Bryant *et al.*, 1998; Bisson *et al.*, 2004; Foa *et al.*, 2006; Sijbrandij *et al.*, 2007; evidence level 2: Bryant *et al.*, 1999; 2005). In addition, research by Bryant *et al.* (1998) and Echeburua *et al.* (1996) showed that relaxation (i.e. anxiety management such as breathing exercises, muscle relaxation, and "self-talk") does not have any added value to trauma-focused CBT alone (evidence level 4).

Eye Movement Desensitization and Reprocessing (EMDR) is a psychotherapeutic procedure that is aimed explicitly at the treatment of PTSD and other anxiety disorders after shocking events. Notwithstanding the accumulating evidence of EMDR as an effective treatment for chronic PTSD (e.g. Maxfield & Hyer, 2002; Seidler & Wagner, 2006), as yet there is insufficient scientific support for the use of EMDR during the first 6 weeks after a stressful life event (evidence level 3: Silver *et al.*, 2005).

Research is also inconclusive about pharmacotherapy as an early intervention, although experts are of the opinion that pharmacotherapy might be considered during the acute phase of PTSD for the treatment of sleep disorders (evidence level 4: NICE, 2005). Furthermore, there are indications that benzodiazepines are not indicated for the treatment of ASD (evidence level 3: Gelpin *et al.*, 1996). For the treatment of depression and anxiety disorders by using medication, The Panel refers to existing guidelines.

Workplace-focused interventions are interventions that take place after shocking events that occur within the workplace. There are indications that workplace-focused interventions result in a significant reduction in psychological problems (evidence level 3: Boscarino *et al.*, 2005), but it remains unclear which workplace-focused interventions are most effective (evidence level 4: Devilly & Cotton, 2003). The Panel recommends that employers should offer counseling after a shocking event takes place at work.

The early curative interventions for children with clinically evident problems should consist of a number of subsequent sessions (evidence level 3: de Roos & Eland, 2005). Although more research is needed, there are indications that early trauma-focused CBT is an effective treatment method for severe symptoms of PTSD in children > 7 years of age (evidence level 4). There is no empirical evidence for the use of EMDR as an early curative intervention in children and adolescents (evidence level 4). The Panel considers that the evidence on the use of pharmacotherapy among children is too limited to recommend it.

The ethnic minorities that are affected by a disaster should be approached as regularly as possible and as culture-specifically as necessary. The victims from different cultures are similar in their expression of the symptoms that are associated with PTSD, and stress and anxiety disorders (evidence level 3). In the opinion of The Panel, culture-specific elements

consist of giving information in the victim's mother tongue and involving key figures from the ethnic minority group.

### Organization (recommendations 32–36)

Psychosocial care after disasters should be integrated within existing procedures on crisis management and it should be offered as regularly as possible. Although there are indications that early interventions should be carried out by trained caregivers (evidence level 4: Arendt & Elklit, 2001), given the diversity of these caregivers (including volunteers, general practitioners, social workers, and the emergency services personnel who are deployed after a shocking event), Parker *et al.* (2006) argued that it is unrealistic to train them in the whole range of early interventions. Instead, they suggested a number of evidence-informed competencies that “public health workers” should satisfy: (i) listen actively; (ii) prioritize and react to the needs of those affected; (iii) recognize minor psychological problems and give information about them; (iv) recognize potentially serious psychological problems and give information about them; (v) communicate the techniques for dealing with ASD; (vi) recognize the risk factors of a poor outcome for mental health and reduce these risk factors through greater awareness; (vii) recognize and use informal and formal resources for interpersonal support; and (viii) refer the victim for more formal forms of mental health care.

The Panel recommends setting up an information and advice center in the week after a disaster. Its main activities are to: (i) reach out to and keep in contact with all those affected (outreach); (ii) collect all the questions of those affected; (iii) formulate answers to these questions; (iv) monitor the condition of those affected; and (v) give advice about the help that the victims seek. In addition, the self-organization of the victims should be encouraged and facilitated. In certain postdisaster situations, it is recommended that specialist mental health-care teams are organized, consisting of people trained in specific forms of treatment based on the needs of the victims. By having the relief workers involved no more than 2–3 days per week, the secondary traumatization of the relief workers and the “monotonousness” of the work itself can be avoided. Finally, health assessments are needed to monitor the health of those affected and, based on these assessments, support can be directed appropriately.

## DISCUSSION

These guidelines for early psychosocial interventions provide, for the first time in the Netherlands, a national standard of what to do during the first 6 weeks after a disaster, terrorist attack, or other shocking event. They can be used as a basis for (local and national) policy and protocols. Notwithstanding the extensive literature search that was carried out, a sizable number of recommendations are primarily based on the opinions of, and consensus among, health-care workers, combined in a national panel of experts. This lack of scientific founding, however, should not be perceived as a limitation. As we pointed out, the professional community is in dire

need of clarity. The current guidelines alleviate this need by providing a standard to which relevant national organizations have committed themselves. In addition, the Dutch guidelines include a “research agenda” by pinpointing those topics in current practice that need further scientific investigation. Regrettably, most of the literature to date focuses on PTSD as the most relevant outcome. The numerous other possible outcomes of witnessing a shocking event receive much less attention; that is, depressive disorders, anxiety disorders, unexplained physical complaints, and substance abuse. Finally, no guidelines are ever meant to serve as a permanent gold standard. Instead, they are to be updated and refined when this is called for by new insights and new scientific evidence.

The need to reduce variation and to upgrade the quality of psychosocial care, using the strongest evidence available, is recognized internationally. In 2007, the Australian Centre for Posttraumatic Mental Health (2007) published the *Australian Guidelines for the Treatment of Adults with Acute Stress Disorder and Posttraumatic Stress Disorder*. In 2005, the PTSD guidelines of NICE were presented (NICE, 2005). A systematic comparison of independently developed guidelines would be interesting. A high level of similarity in the recommended treatment and support interventions would imply that the situation in different countries is largely generic when it comes to the psychosocial needs of victims and the applicability and appropriateness of psychosocial care interventions. The effective implementation of the guidelines in their respective country then would lead automatically to a reduction of international variation. Interestingly, in a first attempt to make such a comparison, Taal (2008) indeed found a large overlap in the scientific literature and conclusions between the Australian and the Dutch guidelines. In the same vein, The European Network for Traumatic Stress has attempted to systematically review the evidence regarding post-traumatic stress management for victims of natural and other disasters (The European Network for Traumatic Stress, 2009).

The extent to which guidelines are capable of meeting their objective of reducing variation in the care that is delivered ultimately depends on the willingness of care workers, professionals, and volunteers to comply with the recommendations. As soon as these groups do not experience guideline ownership, or the purpose, applicability, or contents of the guidelines are scrutinized in any way, the prospects of guideline compliance are minimal. The Dutch guidelines were accepted by all but one of the 18 key organizations listed in Appendix I. This ensures the legitimacy of the guidelines in the field. Moreover, recommended interventions must have a clear advantage, be relatively simple to understand, straightforward to apply, and – however paradoxically this might seem – must not interfere too heavily with existing practices (Greenhalgh *et al.*, 2004).

In their description of the available evidence for intervention policies in immediate and mid-term mass trauma interventions, Hobfoll *et al.* (2007: 283) noted that “. . . to date, no evidence-based consensus has been reached supporting a clear set of recommendations”. To address this issue, the authors employed an “evidence-informed” approach to the

subject by extrapolating the findings from related fields of research and to gain consensus on the intervention principles. Five "essential elements" were identified: (i) a sense of safety; (ii) a sense of calm; (iii) self-efficacy and community efficacy; (iv) a sense of connectedness; and (v) hope (Hobfoll *et al.*, 2007). In general, these five principles are in correspondence with the Dutch guidelines, specifically concerning the principle of the provision of a supportive context. Although this principle is not endorsed by strong empirical evidence, we concur with Hobfoll *et al.* that it is unlikely that clinical trials that cover the diversity of disaster situations are feasible. As a healthy psychosocial practice requires some kind of standard on what to do, we believe that consensus among experts, based on intuitive and straightforward thinking, is the most reasonable alternative.

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- Dutch Association for Psychotherapy (Nederlandse Vereniging voor Psychotherapie)
- Netherlands Institute of Psychologists (Nederlands Instituut van Psychologen)
- Dutch College of General Practitioners (Nederlands Huisartsen Genootschap)
- Dutch Association of Primary Care Psychologists (Landelijke Vereniging van Eerstelijnspsychologen)
- Military Mental Health Care Institute of the Ministry of Defence (Militaire Geestelijke Gezondheidszorg)
- Netherlands Association of Policy, Management and Research Physicians (Nederlandse Vereniging Artsen Beleid Management Onderzoek)
- Netherlands Association of Social Workers (Nederlandse Vereniging van Maatschappelijk Werkers)
- Netherlands Association of Fire and Disaster Control Services (Nederlandse Vereniging voor Brandweezorg and Rampenbestrijding)
- Netherlands Society of Physicians in Occupational Health (Nederlandse Vereniging voor Arbeids – en Bedrijfsgeneeskunde)
- Dutch Association of Behavioral and Cognitive Therapy (Vereniging voor Gedragstherapie en Cognitieve Therapie)
- Institute for Psychotrauma (Instituut voor Psychotrauma)
- Immediate Relief and Aftercare (Directe Opvang en Nazorg)
- Victim Support (Slachtofferhulp Nederland; formerly National Victim Support Agency)
- Council of Regional Medical Officers (Raad van Regionaal Geneeskundig Functionarissen)
- Regional Department of Emergency and Disaster Medicine Preparedness (Landelijk Bureau Geneeskundige Hulpverlening na Ongevallen en Rampen)
- Police Academy of the Netherlands (Politieacademie)

## APPENDIX I

### Cooperating professional associations and organizations that were represented in the national, multidisciplinary panel of experts that developed the guidelines on early psychosocial interventions after disasters, terrorism, or other shocking events

Mental Healthcare Nursing Federation (Federatie Verpleegkunde in de Geestelijke Gezondheidszorg)